Group Project: Part 2

Question 1: How many variables are there?

Our dataset contains six variables. The first variable, Year, identifies the respondent's current level of study, such as Freshman, Sophomore, Junior, Senior, Graduate Student, or PhD. This allows us to see whether academic standing influences learning preferences.

The second variable, Commute, describes how students travel to campus. The options included Walk, Drive personal vehicle, Public transportation, Bicycle/Scooter, Campus shuttle, and Other. This variable helps determine whether travel methods affect attendance when weather conditions are poor.

The third variable, Weather_Impact_Level, represents how strongly severe weather conditions (such as snow, ice, or extreme temperatures) influence a student's ability or willingness to attend in-person classes. Responses were recorded on a five-point scale.

The fourth variable, Class_Preference, shows whether a student prefers online or in-person classes, ranging from "Strongly prefer online" to "Strongly prefer in-person." This is the main variable used to test our hypothesis.

The fifth variable, Internet_Access, indicates how reliable a student's internet connection is at home, which is important for assessing readiness for remote learning.

Finally, Home_Comfort reflects how comfortable and quiet the student's home environment is for studying or attending online classes. Together, these six variables capture the main factors influencing whether students would prefer online or in-person learning under different weather conditions.

Question 2: How many observations are there?

There are thirteen observations in the dataset, representing thirteen individual students. Each record includes complete data for all six variables, providing a consistent and analyzable dataset.

Question 3: How did you define the number of data points you need?

The number of data points was defined based on project feasibility and scope. Initially, the plan was to gather about thirty to forty responses from Stevens Institute of Technology students. This target was expected to provide enough variation for analysis while remaining manageable for a student project. However, due to limited participation time and voluntary response, the final dataset includes thirteen complete responses. Although smaller than planned, this number is adequate to identify key patterns and demonstrate statistical techniques related to the study hypothesis.

Question 4: What sources of bias came up as you collected the data?

Three main types of bias were identified during data collection. The first was Sampling Bias, because most responses were collected from Stevens Institute of Technology students. This overrepresentation of a single institution means the findings may not generalize well to students from other universities or regions with different commuting conditions and weather experiences.

The second was Voluntary Response Bias, since participation was optional and students who had stronger opinions about weather or online learning may have been more likely to respond.

The third was Interpretation Bias, because participants might define “severe weather” differently based on their personal experiences and tolerance.

To reduce sampling bias, the survey was later shared with students from other colleges to include a wider range of perspectives.